# ASSIGNMENT 5

## Database Creation and Table Implementation

**Project: Online Shopping Management System**

## Aim / Objective

Create OnlineShoppingDB and implement the five tables exactly as defined in your uploaded SQL file.

## 1. Complete database and table creation code

```
CREATE DATABASE OnlineShoppingDB;
USE OnlineShoppingDB;

CREATE TABLE Customers (
 CustomerID INT PRIMARY KEY AUTO_INCREMENT,
 CustomerName VARCHAR(100) NOT NULL,
 Email VARCHAR(100) NOT NULL UNIQUE,
 Phone VARCHAR(15),
 Address VARCHAR(255)
);
CREATE TABLE Products (
 ProductID INT PRIMARY KEY AUTO_INCREMENT,
 ProductName VARCHAR(100) NOT NULL,
 Category VARCHAR(50) NOT NULL,
 Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
 Stock INT NOT NULL DEFAULT 0 CHECK (Stock >= 0)
);
CREATE TABLE Orders (
 OrderID INT PRIMARY KEY AUTO_INCREMENT,
 CustomerID INT NOT NULL,
 OrderDate DATE NOT NULL,
 Status VARCHAR(20) NOT NULL DEFAULT 'Pending',
 FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);
CREATE TABLE OrderDetails (
 OrderDetailID INT PRIMARY KEY AUTO_INCREMENT,
 OrderID INT NOT NULL,
 ProductID INT NOT NULL,
 Quantity INT NOT NULL CHECK (Quantity > 0),
 UnitPrice DECIMAL(10,2) NOT NULL CHECK (UnitPrice > 0),
 FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
 FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);
CREATE TABLE Payments (
 PaymentID INT PRIMARY KEY AUTO_INCREMENT,
 OrderID INT NOT NULL,
 PaymentMethod VARCHAR(30) NOT NULL,
 Amount DECIMAL(10,2) NOT NULL CHECK (Amount > 0),
 PaymentStatus VARCHAR(20) NOT NULL DEFAULT 'Pending',
 FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)
);
```

# 2. Output

**TABLE OF CUSTOMER**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| CustomerID | int | NO | PRI | NULL | auto_increment |
| CustomerName | varchar(100) | NO | | NULL | |
| Email | varchar(100) | NO | UNI | NULL | |
| Phone | varchar(15) | YES | | NULL | |
| Address | varchar(255) | YES | | NULL | |

**TABLE OF PRODUCTS**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| ProductID | int | NO | PRI | NULL | auto_increment |
| ProductName | varchar(100) | NO | | NULL | |
| Category | varchar(50) | NO | | NULL | |
| Price | decimal(10,2) | NO | | NULL | |
| Stock | int | NO | | 0 | |

**TABLE OF ORDERS**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| OrderID | int | NO | PRI | NULL | auto_increment |
| CustomerID | int | NO | MUL | NULL | |
| OrderDate | date | NO | | NULL | |
| Status | varchar(20) | NO | | Pending | |

**TABLE OF ORDER DETALIS**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| OrderDetailID | int | NO | PRI | NULL | auto_increment |
| OrderID | int | NO | MUL | NULL | |
| ProductID | int | NO | MUL | NULL | |
| Quantity | int | NO | | NULL | |
| UnitPrice | decimal(10,2) | NO | | NULL | |

**TABLE OF PAYMENTS**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| PaymentID | int | NO | PRI | NULL | auto_increment |
| OrderID | int | NO | MUL | NULL | |
| PaymentMethod | varchar(30) | NO | | NULL | |
| Amount | decimal(10,2) | NO | | NULL | |
| PaymentStatus | varchar(20) | NO | | Pending | |

# 4. Conclusion

The schema defines five linked tables with primary keys, foreign keys, required values, a unique email, defaults, and CHECK constraints.